

Roi Bein

[Portfolio](#) | rb505@duke.edu | (864) 650-7422

EDUCATION

Duke University, Durham, NC
BSE - Mechanical Engineering
Certificate: Robotics & Automation
GPA: 3.96

Expected May 2029

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Onshape, CFD, FEA
Manufacturing: CNC Mill, Lathe, Waterjet, Laser Cutter, 3D Printing
Programming: Python, Java, JavaScript, C, SwiftUI, Arduino
Software: Microsoft Office, Photoshop
Languages: English, Spanish, Hebrew

WORK EXPERIENCE

- Duke Co-Lab, Makerspace Technician** January 2026-Present
- Train students and researchers on equipment including 3D printers, laser cutters, waterjet, and ShopBot CNC systems
 - Maintain 45+ 3D printers, perform troubleshooting, repairs, and preventative maintenance
- Duke University Luck Biomechanics Lab, Research Assistant** January 2026-Present
- Conduct biomechanics experiments using a drop tower, load cell instrumentation, and anthropomorphic test devices
 - Analyze and graph 600+ trials of gyroscopic acceleration data using Python
- Duke University, Software Engineering Intern** May-July 2026
- Developed full-stack functionality for a research mobile application used by 100+ participants to collect wearable health data, building SwiftUI frontend interfaces and FastAPI backend services
 - Built fitness dashboards, Duke Recreation event integration, social leaderboards, and in-app games to improve user engagement and support continuous health-data collection
- Camp Judaea, counselor (zipline and climbing wall specialist)** July 2025
- Oversaw zipline and climbing wall activities for campers while maintaining equipment and enforcing safety protocols
- Clemson University Department of Bioengineering, Lab Aide Intern** May-August 2024
- Created and passaged cell cultures used for testing
 - Designed and developed an Arduino-controlled microscope stage using stepper motors and CAD-designed components
- Army Educational Outreach Program (AEOP) Internship, Lab Aide Intern** June-August 2023
- Created cell cultures, ran experiments, and gathered data on cell signals with a flow cytometer
 - Trained machine learning models on this data to recognize differences between cell signals
 - Contributed to research published in *IEEE Sensors Journal* ([DOI: 10.1109/JSEN.2025.3551914](#))

PROJECTS

- Aquatic Knee Stabilization Device** August - December 2025
- Collaborated to create a knee brace that functions underwater to aid a spinal cord injury patient in physical therapy
 - Used CAD and physical/mechanical/iterative design to engineer a working prototype
- Microscope Stage Adjuster** May-August 2024
- Created a stage that can move precisely in 3 axes using Arduino and stepper motors
- Personal Projects** (see [portfolio](#))
- CD Display Case, Happy Meal Toy, Sunflower Sheller, Toy RoboCop Helmet
 - CAD projects demonstrating tolerance analysis, DFM principles, and iterative mechanical design

EXTRACURRICULAR EXPERIENCE

- Duke Motorsports Team, Vehicle Dynamics** August 2025-Present
- Collaborate with 20+ engineers to design, manufacture, and test a Formula SAE race car
 - Design and validate Formula SAE components using CAD, CFD, and FEA prior to fabrication
 - Machine and manufacture parts using CNC mill, lathe, and 3D printers
- Duke STEM Connect** October 2025-Present
- Volunteer at a local Durham elementary school to design and teach fun and engaging STEM lessons
 - Build relationships with 20+ students and encourage engagement with STEM subjects

INTERESTS

Basketball, cinema, Brazilian Jiu-Jitsu, music making, chess, old school hip-hop, Rubik's Cubes